

Part III

ENGINEERING PRACTICES

before we get start

We use claude code as our AI coding tools, and use vscode as our code editor, and use git as our version control system, and use github as our code hosting platform. We need to set them up before we get start.

git setup

Let's firstly set up git, cause the cluade code needs git bash to run. steps:

- go to <https://git-scm.com/install/windows>
- download the git for windows installer, and run it, and follow the instructions to install it
- for the components selection part, check everything, and the others can be default, and click next

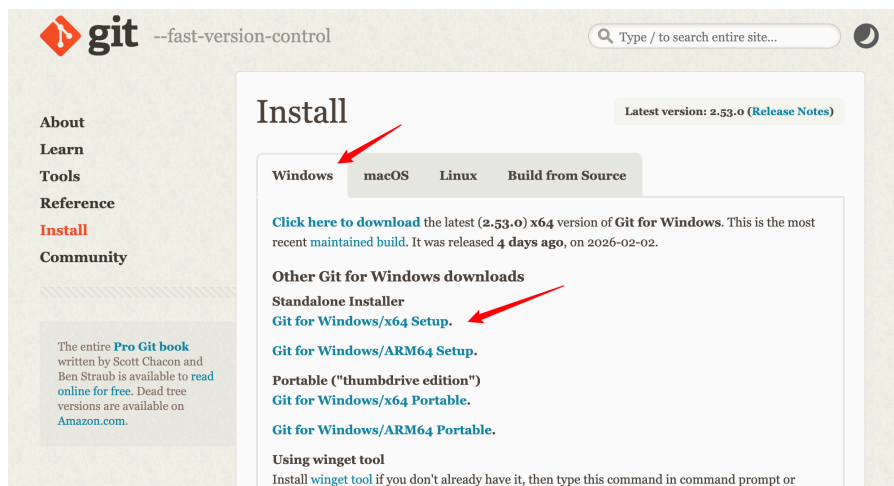


Figure 7.1: Installing Git on Windows

Select Components

Which components should be installed?

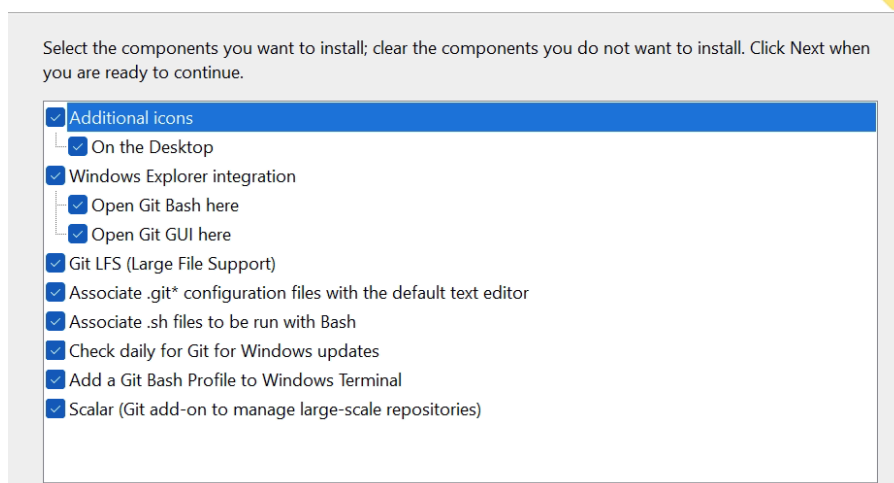


Figure 7.2: check everything in this step

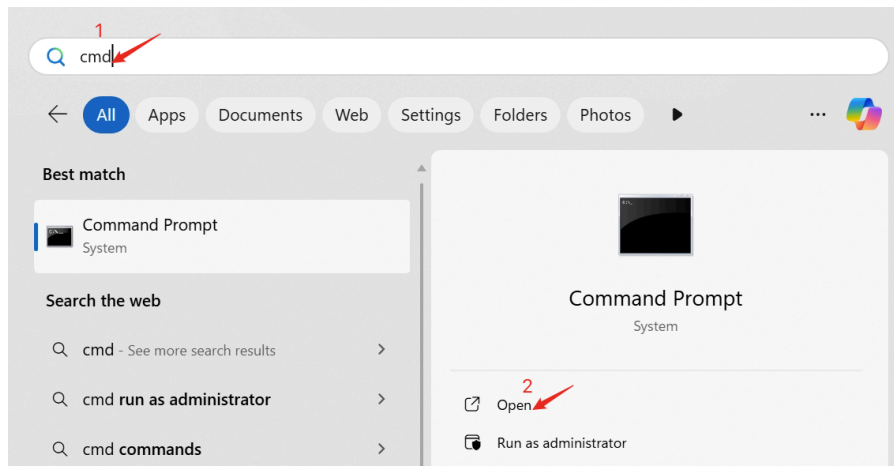


Figure 7.3: open the command prompt, and run "git --version" to check if git is installed successfully

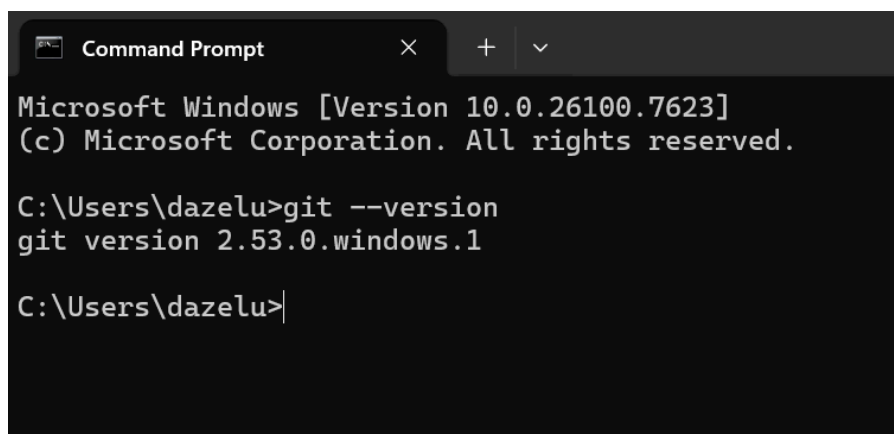


Figure 7.4: if you see this, it means you have successfully installed git

nodejs setup

To build up a web application, we need to set up nodejs environment, please go to <https://nodejs.org/en/download> and download the latest LTS version of nodejs, and run the installer, and follow the instructions to install it. If you have administration permission, it will be easy, if you don't have administration permission, we will install nodejs manually, in our case, we will do it manually.

steps:

- go to <https://nodejs.org/en/download>, and download the latest LTS version of nodejs for windows, and choose the binary zip version, not the installer version.
- after downloading, extract the zip file to a folder, for example, "D:\nodejs"
- add the nodejs folder to the PATH environment variable, so that we can run node and npm commands in the terminal, to do this, search "environment variable" in the windows search bar, and click "Edit the system environment variables", then click "Environment Variables", then find the "Path" variable in the "user variables" section, and click "Edit", then click "New", and add the path of the nodejs folder, for example, "D:\nodejs", then click "OK" to save the changes.
- open a new command prompt, and run "node -v" and "npm -v" to check if nodejs is installed successfully, if you see the version numbers, it means you have successfully installed nodejs.

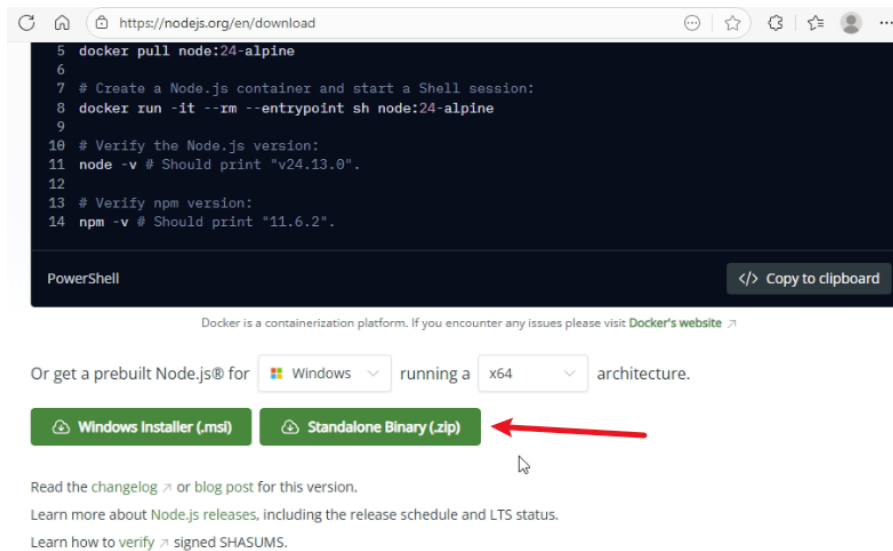


Figure 7.5: download nodejs binary zip file

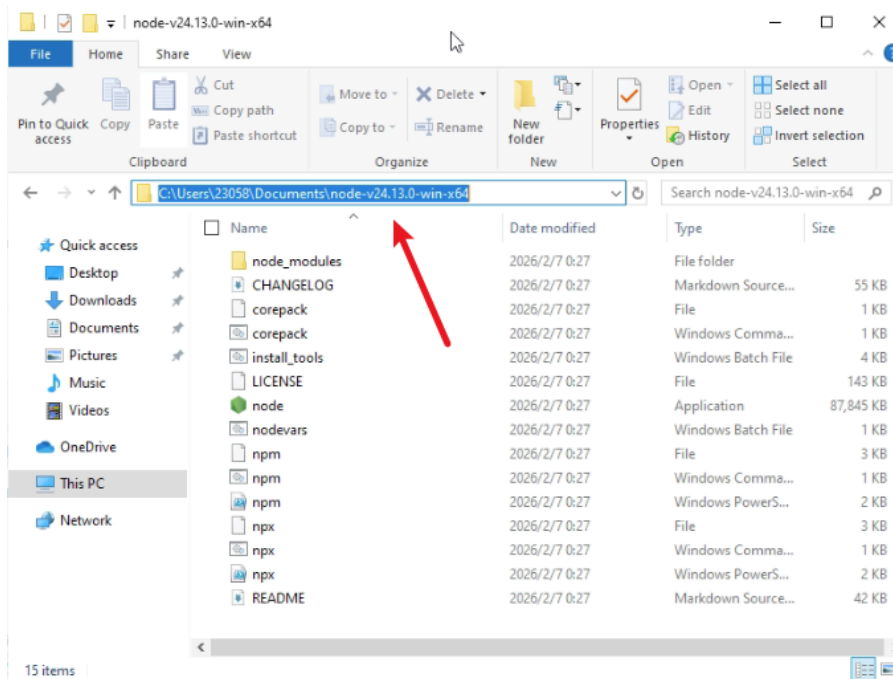


Figure 7.6: extract the nodejs zip file to a folder

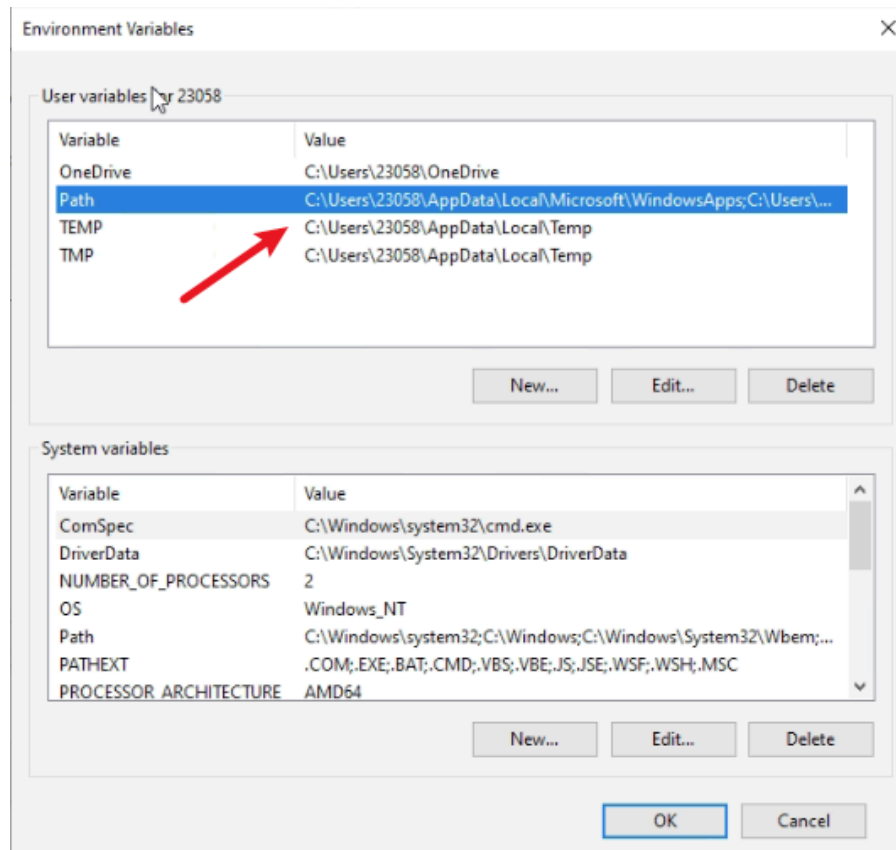


Figure 7.7: add nodejs to the PATH environment variable, click the edit button to add new path

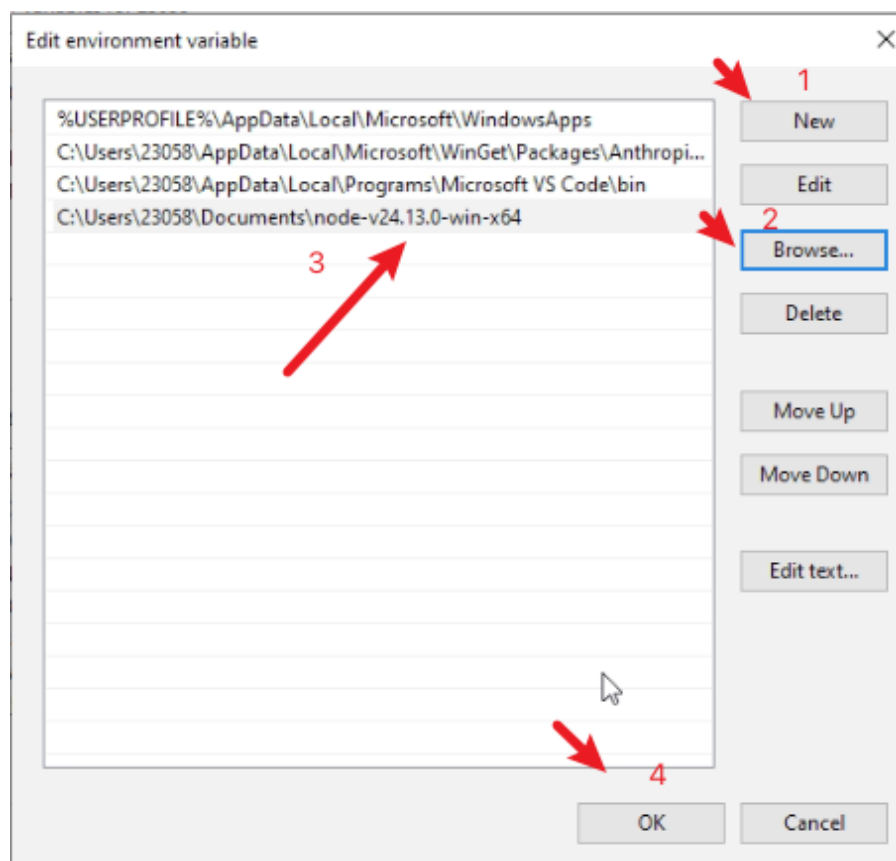
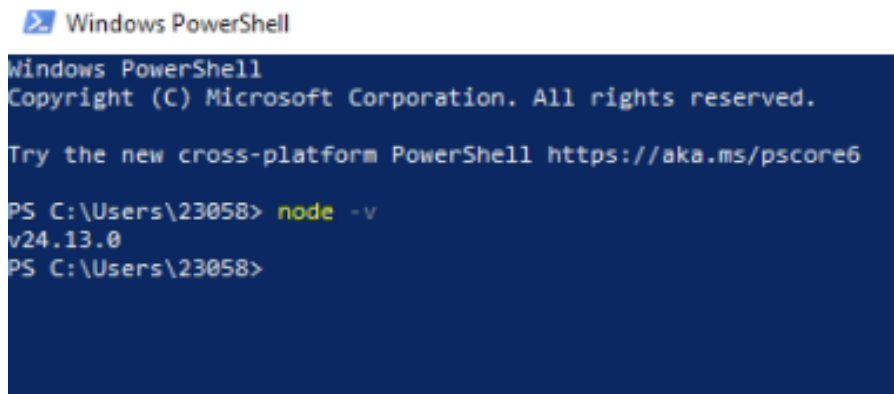


Figure 7.8: click new and add navigate to the nodejs folder to add it to the PATH environment variable



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

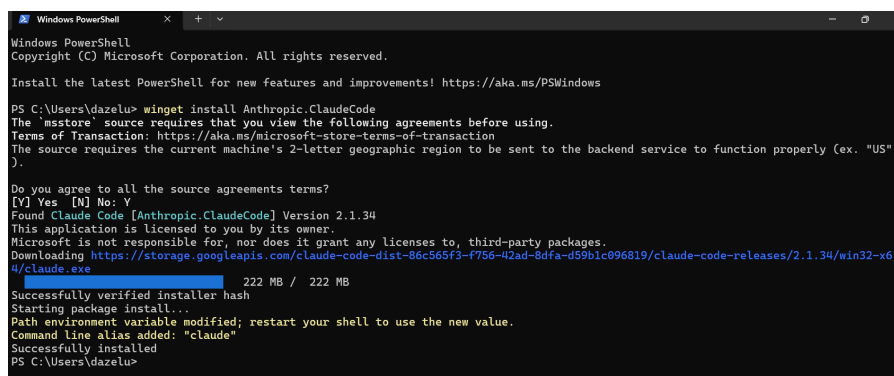
PS C:\Users\23058> node -v
v24.13.0
PS C:\Users\23058>
```

Figure 7.9: nodejs is successfully installed and added to PATH environment variable if you can see the version numbers after running "node -v" and "npm -v" in a new powershell or command prompt

claude code setup

Go to <https://code.claude.com/docs/en/overview>, you can find the installation instructions for different platforms. For example, for Windows, you can run the following command in PowerShell to install claude code, to find the powershell, in the top of the command prompt, click the down arrow, and select "Windows PowerShell", then run the command below:

```
# Windows (Winget):
winget install Anthropic.ClaudeCode
```

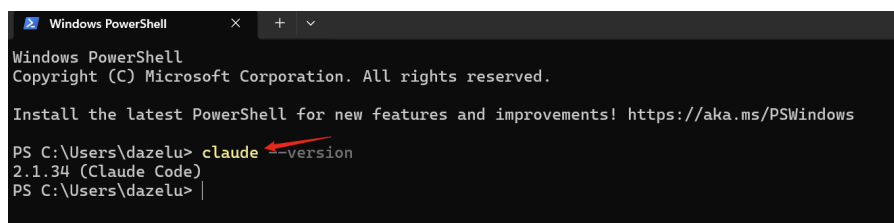


```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\dazelu> winget install Anthropic.ClaudeCode
The 'msstore' source requires that you view the following agreements before using.
Terms of Transaction: https://aka.ms/microsoft-store-terms-of-transaction
The source requires the current machine's 2-letter geographic region to be sent to the backend service to function properly (ex. "US").
Do you agree to all the source agreements terms?
[Y] Yes [N] No: Y
Found Claude Code [Anthropic.ClaudeCode] Version 2.1.34
This application is licensed to you by its owner.
Microsoft is not responsible for, nor does it grant any licenses to, third-party packages.
Downloading https://storage.googleapis.com/claude-code-dist-86c565f3-f756-42ad-8dfa-d59b1c096819/claude-code-releases/2.1.34/win32-x64/claude.exe
222 MB / 222 MB
Successfully verified installer hash
Starting package install...
Path environment variable modified; restart your shell to use the new value.
Command line alias added: "claude"
Successfully installed
PS C:\Users\dazelu>
```

Figure 7.10: if you see this, it means you have successfully installed claude code



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\dazelu> claude --version
2.1.34 (Claude Code)
PS C:\Users\dazelu>
```

Figure 7.11: check the version in a new powershell, if you see the version number, it means you have successfully installed claude code

Next step, we need to setup the API key for claude code, we use z.ai API in claude code, because it's cheaper than the claude API. Go to <https://z.ai/>, and sign up for an account, then go to the API section, and create a new API key for claude, see this link for guide: <https://docs.z.ai/devpack/tool/claude>.

In this class, I'll provide my API-KEY for you to use, run these commands in the gitbash to set up the API key for claude code:

```
curl -O "https://cdn.bigmodel.cn/install/claude_code_zai_env.sh" && bash ./claude_code_zai_env.sh
```

then paste the following api-key

```
861e62ff738e40439f1e15e33f6e5a37.0pVFAhFjfMmhFrvG
```

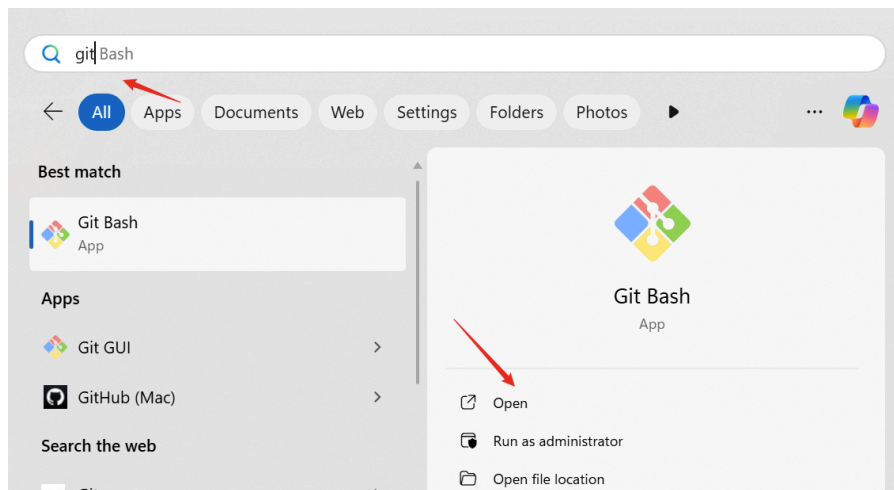


Figure 7.12: find gitbash

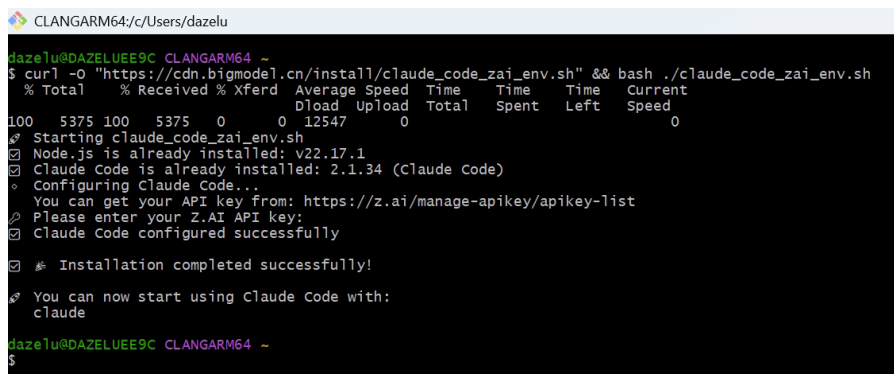


Figure 7.13: set up the API key for claude code in gitbash

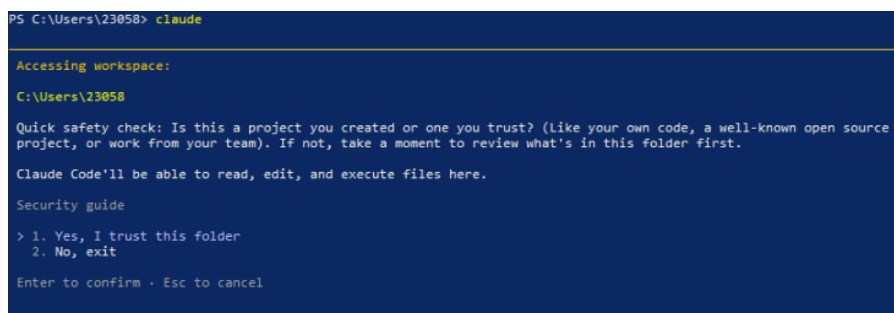


Figure 7.14: if you see this, it means you have successfully set up the API key for claude code, and you can start using it

By the way, if the claude code couldn't be setup, you can use cursor with this API-KEY, please follow this link for guidance: <https://docs.z.ai/devpack/tool/cursor>. And please download cursor from <https://cursor.com/download>.

Python Environment Setup

We use Miniconda to manage the Python environment, please go to <https://repo.anaconda.com/miniconda/> and download the latest version of Miniconda for Windows, remember check "add to path" option when installing it, after you have installed Miniconda, you can open the Anaconda Prompt, and run "conda -version" to check if conda is installed successfully, if you see the version number, it means you have successfully installed Miniconda.

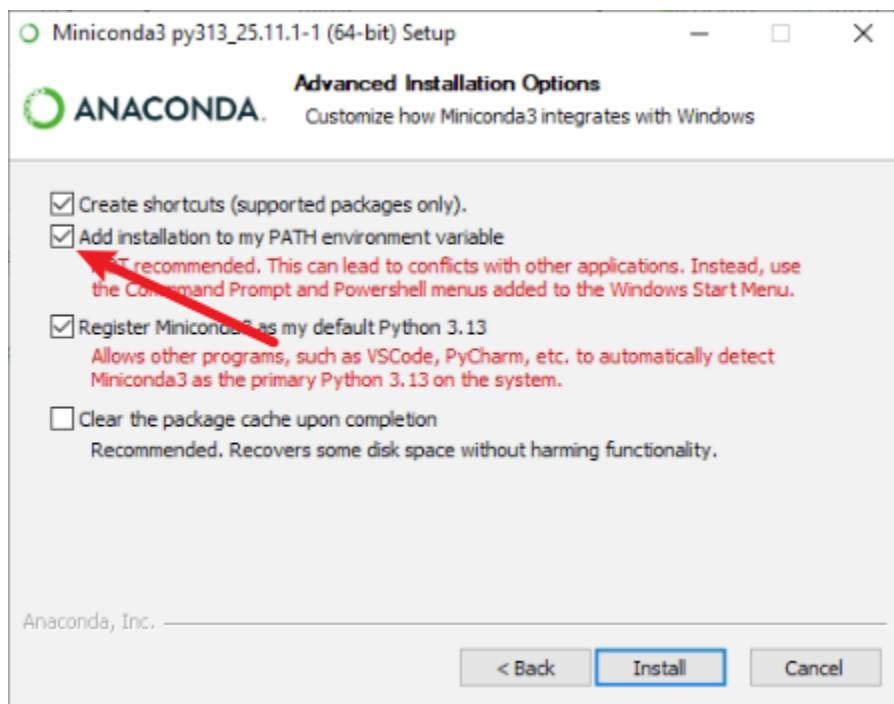


Figure 7.15: Add the conda to the PATH environment variable when installing Miniconda

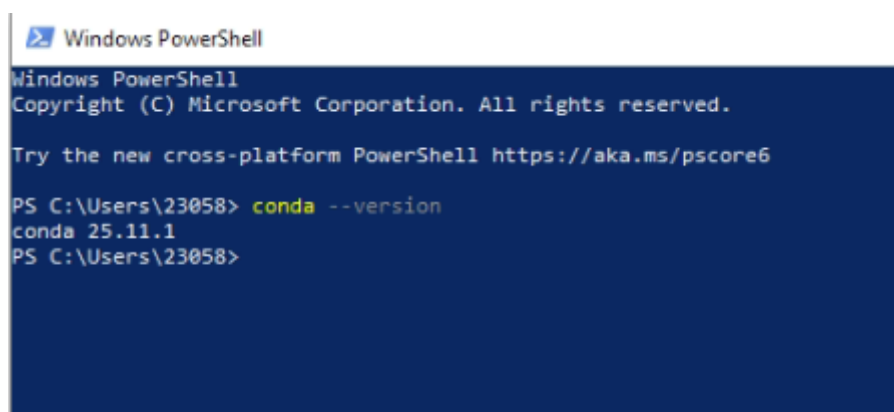


Figure 7.16: Check if install conda successfully by running "conda -version" in a new Anaconda Prompt, if you see the version number, it means you have successfully installed Miniconda

vscode setup

It's easy to set up vscode, please download from <https://code.visualstudio.com/>, you can just choose the default installation options. After you have installed vscode, you can open the folder of your project in vscode, and you can see the code in vscode, and you can also use the terminal in vscode to run git commands, and run the code.

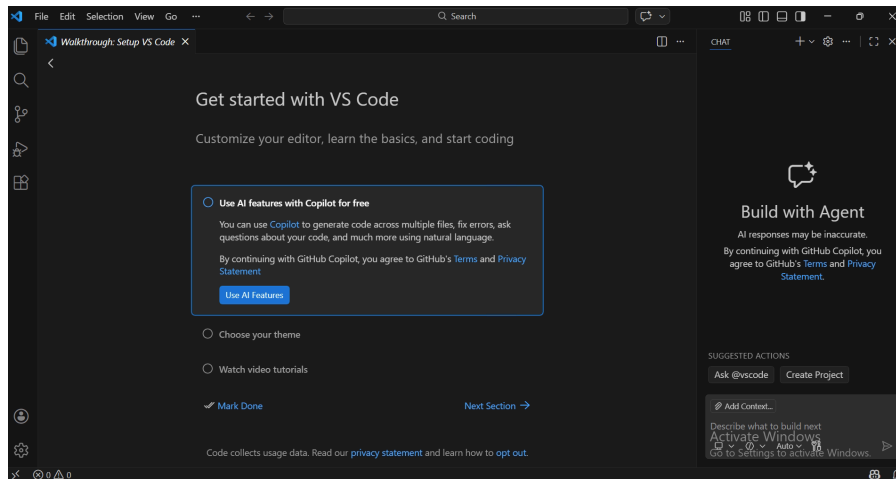


Figure 7.17: you can get page like this after you installing vscode and launching it for the first time

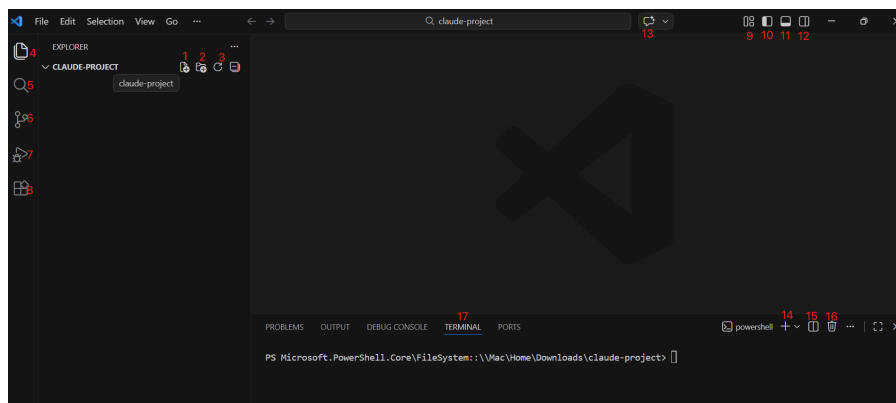


Figure 7.18: vscode introduction

Here is simple introduction for the vscode interface:

- **1 New File:** Create a new file in the current folder.
- **2 New Folder:** Create a new folder in the current folder.
- **3 Refresh:** Refresh the file explorer to show the latest changes.
- **4 Explorer Panel:** Shows the project files and folders.
- **5 Search:** Search text across the project.
- **6 Source Control:** Manage Git version control.
- **7 Run and Debug:** Run or debug the project.
- **8 Extensions:** Install and manage VS Code extensions.
- **9 Tools Bar:** Access various tools and commands.
- **10 Left Panel Controller:** Show/hide the left panel.
- **11 Bottom Panel Controller:** Show/hide the bottom panel.

-
- **12 Right Panel Controller:** Show/hide the right panel.
 - **13 AI Calling:** Call AI assistant in VS Code.
 - **14 New Terminal:** Create a new terminal.
 - **15 Split Terminal:** Split the terminal window.
 - **16 Close Terminal:** Close the current terminal.
 - **17 Terminal Panel:** Integrated terminal for command execution.

github setup

Login hugub with your google account, we will use it to store your code.

Chapter 9 Web Development

9.1 Blog System

System Overview In this project, we develop a blog service system. The backend is implemented using Python and Flask to provide RESTful APIs, with Python's built-in database (SQLite) used for data persistence. The system supports the following functionalities:

- User registration, login, and logout.
- Creation, editing, and publishing of new blog posts.
- Modification and deletion of already published blog posts.
- Support for multiple page themes, such as light mode and dark mode.
- Blog interaction features, including likes, dislikes and comments.
- Blog search functionality.

Finally, we will deploy the blog system through cloudflare, then people can access it publicly.

9.1.1 Product Requirements Document

Now we gonna use AI to give up a plan. Get into claude, and enable plan mode, then input the our system overview, then you just negotiate with the AI to give up a plan for the implementation of the blog system, you can ask the AI to break down the system into smaller components, and give up a plan for each component, and also give up a timeline for the implementation.

```
/model to try Opus 4.5
//plan
Enabled plan mode

System Overview In this project, we develop a blog service system. The backend is implemented using
Python and Flask to provide RESTful APIs, with Python's built-in database (SQLite) used for data persistence.
The system supports the following functionalities:
User registration, login, and logout.
Creation, editing, and publishing of new blog posts.
Modification and deletion of already published blog posts.
Support for multiple page themes, such as light mode and dark mode.
Blog interaction features, including likes, dislikes and comments.
Blog search functionality

I see you've entered plan mode and provided an overview of your blog service system. However, I don't see a specific task or feature you'd like me to help plan.
Let me ask you to clarify what you'd like to work on:

Task

1. Add a new feature
Implement functionality not currently described (e.g., tags, categories, user profiles, etc.)
2. Improve existing feature
Enhance something the system already does (e.g., better search, theme switching, etc.)
3. Fix a bug or issue
Debug or fix something that's not working correctly
4. Set up from scratch
Help design and plan the initial architecture for this blog system
5. Type something
6. Chat about this
7. Skip interview and plan immediately

Enter to select · r/ to navigate · Esc to cancel
```

Figure 9.1: AI plan mode in claude

 **Note** Please let the AI generate the plan into a file, like 'task.md', you may need to modify it later.

```

Phase 8: Testing & Deployment
- Unit and integration tests
- Docker configuration
- CI/CD pipeline
- Deployment documentation
---
Critical Files
- backend/app/models/post.py - Post model with tag/category relationships
- backend/app/routes/posts.py - Post CRUD with filtering, search, auth
- backend/app/services/auth_service.py - JWT token generation/verification
- backend/app/extensions.py - Flask extensions (db, jwt, cors)
- frontend/src/components/posts/PostEditor.jsx - Toast UI Editor integration
- frontend/src/services/api.js - Axios with JWT interceptors
- frontend/src/contexts/AuthContext.jsx - Authentication state management
---
Verification
1. Backend: Run flask run and test API endpoints with Postman/Thunder Client
2. Frontend: Run npm run dev and verify full user flow (register → login → create post → add tags/category → publish)
3. Interactions: Test like/dislike, comments, nested replies
4. Search: Verify search by keyword, tag filter, category filter
5. Themes: Toggle light/dark mode and verify persistence
6. Database: Check SQLite tables for correct relationships

Claude has written up a plan and is ready to execute. Would you like to proceed?
> 1. Yes, clear context and auto-accept edits (shift+tab)
2. Yes, auto-accept edits
3. Yes, manually approve edits
4. Type here to tell Claude what to change

```

Figure 9.2: AI negotiation mode in claude

After several rounds of negotiation, you can get a plan for the implementation of the blog system, you can accept the plan and proceed with the development or keep changing it until it is satisfactory.

Now we use vscode to open the project folder, and create a new file named 'task.md', then copy the plan generated by AI into this file, and we can refer to this file during the development process.

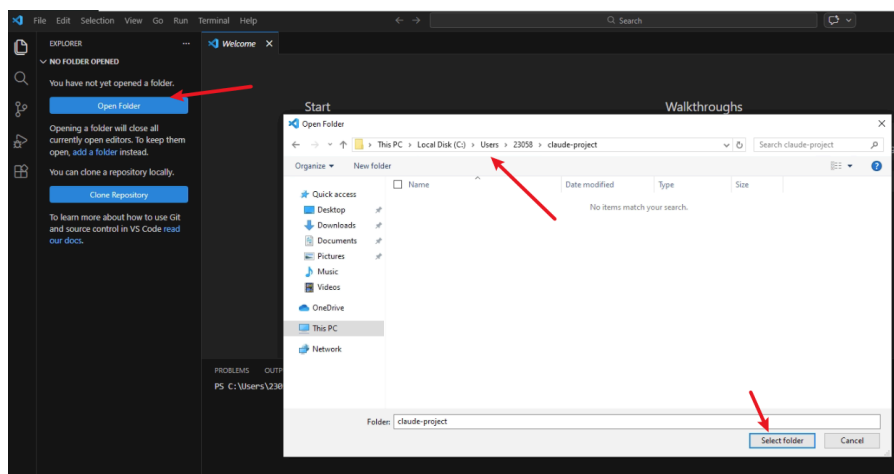


Figure 9.3: Open project folder in VS Code

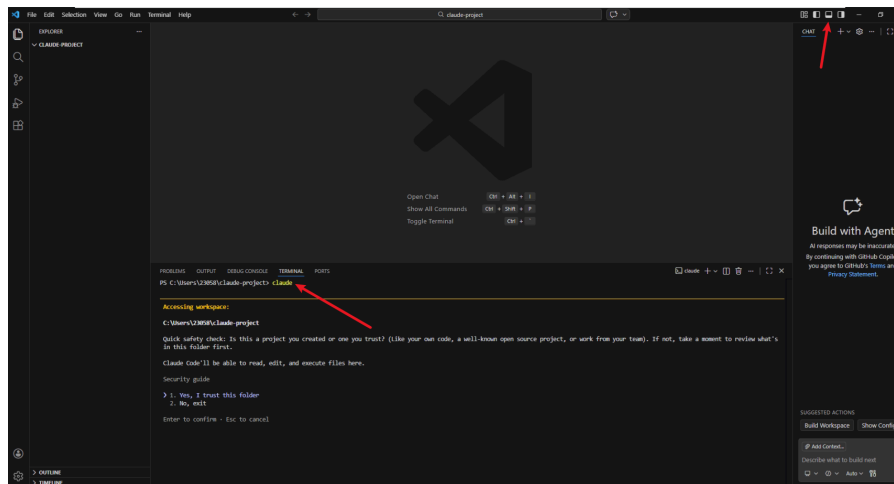


Figure 9.4: Use Claude in VS Code

9.2 Control Code Version and Save on Github

We firstly need to create an empty git repository on github, and copy the remote repository URL, then we will initialize a git repository in our project folder, and add the remote repository to our local git repository, then we can stage all the changes, commit the changes, and push the changes to the remote repository on github.

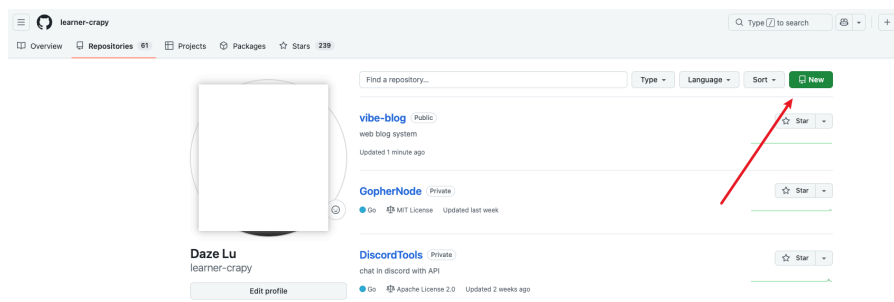


Figure 9.5: create a project on github

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk (*).

- General**

Owner * learner-crapy / Repository name * vibe-blog
✔ vibe-blog is available.

Great repository names are short and memorable. How about [special-octo-invention?](#)

Description web blog system
15 / 350 characters
- Configuration**

Choose visibility * Public
Choose who can see and commit to this repository

Add README On
READMEs can be used as longer descriptions. [About READMEs](#)

Add .gitignore No .gitignore
.gitignore tells git which files not to track. [About ignoring files](#)

Add license No license
Licenses explain how others can use your code. [About licenses](#)

Create repository

Figure 9.6: Create a new git repository on GitHub

 **Note** Please note that your github repo name should be same as your local project folder name.

We then need to initialize a git repository in our project folder. Follow these steps:

1. Initialize a git repository in your project folder by running:

```
git init
```

2. Create a new repository on GitHub and copy the remote repository URL.
3. Add the remote repository to your local git repository:

```
git remote add origin https://github.com/learner-crapy/vibe-blog.git
```

4. Stage all the changes:

```
git add .
```

5. Commit the changes:

```
git commit -m "initial commit"
```

6. Push the changes to the remote repository:

```
git push -u origin master
```

After several rounds of developments, you'll get a system.

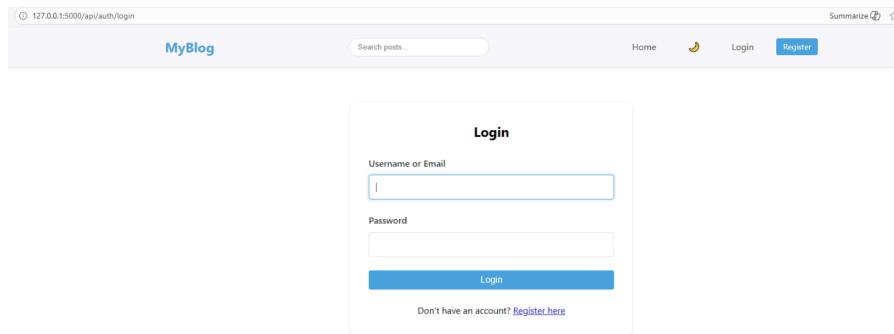


Figure 9.7: the blog system running on local machine

9.3 make your project publicly accessible

After you have implemented the blog system on your local machine, only your local machine can access it. If you want to make it publicly accessible, you need to deploy it to a cloud platform. In this class, we will use Cloudflare to deploy our blog system.

Follow these steps:

1. Download the Cloudflare tunnel client for Windows from:
<https://developers.cloudflare.com/cloudflare-one/networks/connectors/cloudflare-tunnel/downloads/>
2. Start the tunnel by running the following command in your terminal:

```
.\cloudflared.exe tunnel --url http://localhost:5000
```

This will create a public URL for your local blog system that you can share with others.

Windows

Download and install `cloudflared` via [winget](#):

```
winget install --id Cloudflare.cloudflared
```

Alternatively, download the latest release directly:

Type	32-bit	64-bit
Executable	Download	Download

Note

Instances of `cloudflared` do not automatically update on Windows. You will need to perform manual updates.

Figure 9.8: download cloudflare tunnel client for windows

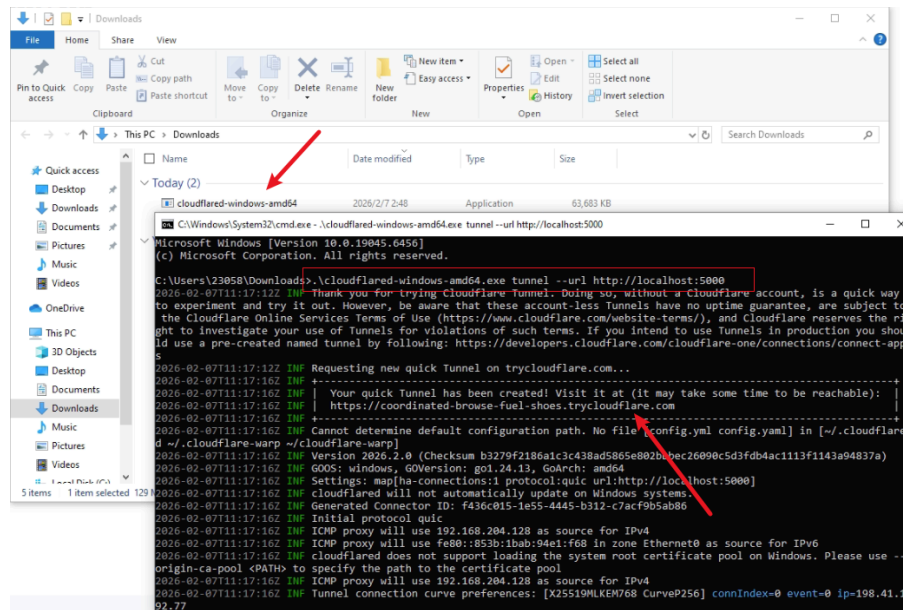


Figure 9.9: cloudflare tunnel client command to start the tunnel and get a public URL for your local blog system